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Introduction to Data Science and Systems

Introduction

Dr Nicolas Pugeault

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GLASGOW

THE SUNDAY TIMES
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UNIVERSITY
GUIDE
2024
SCOTTISH
UNIVERSITY
OF THE YEAR

Outline

1. Introduction & motivation
2. Structure and logistics of the course
3. Getting started with vector spaces





By the end of this week, you should:

- know your lecturer(s) and lab assistant
- know the structure of the course (including assessment and schedule)
- become familiar with common definition(s) of data science
- know the basic steps in data science: load data, identify the data type, perform basic quality control, data cleaning/curation, simple visualizations.
- become familiar with Numpy through the self-study and (supervised) labs





Data is transforming the world!

Technology trends: (adapted from J. Gonzales)

- 1940-1980s: Hardware - building digital computers
- 1990s: Software industry (developing increasingly more complex software)
- 2000s: Internet industry (online retailing and services)
 - *At the same time: Mobile devices*
- 2010s: Data industry (sell, buy and share information about people and objects)
- 2020s: The AI revolution (powered by the availability of large data and parallel hardware)
- 2030s: ???





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Examples: Universities

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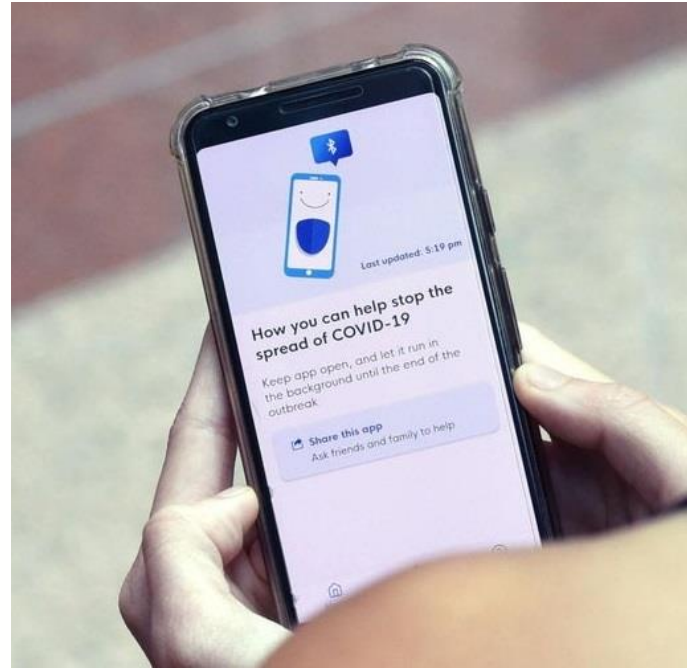
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More examples:

- Politics (CA scandal)
- Dating sites
- Covid tracking
- ...



How was Facebook users' data misused?

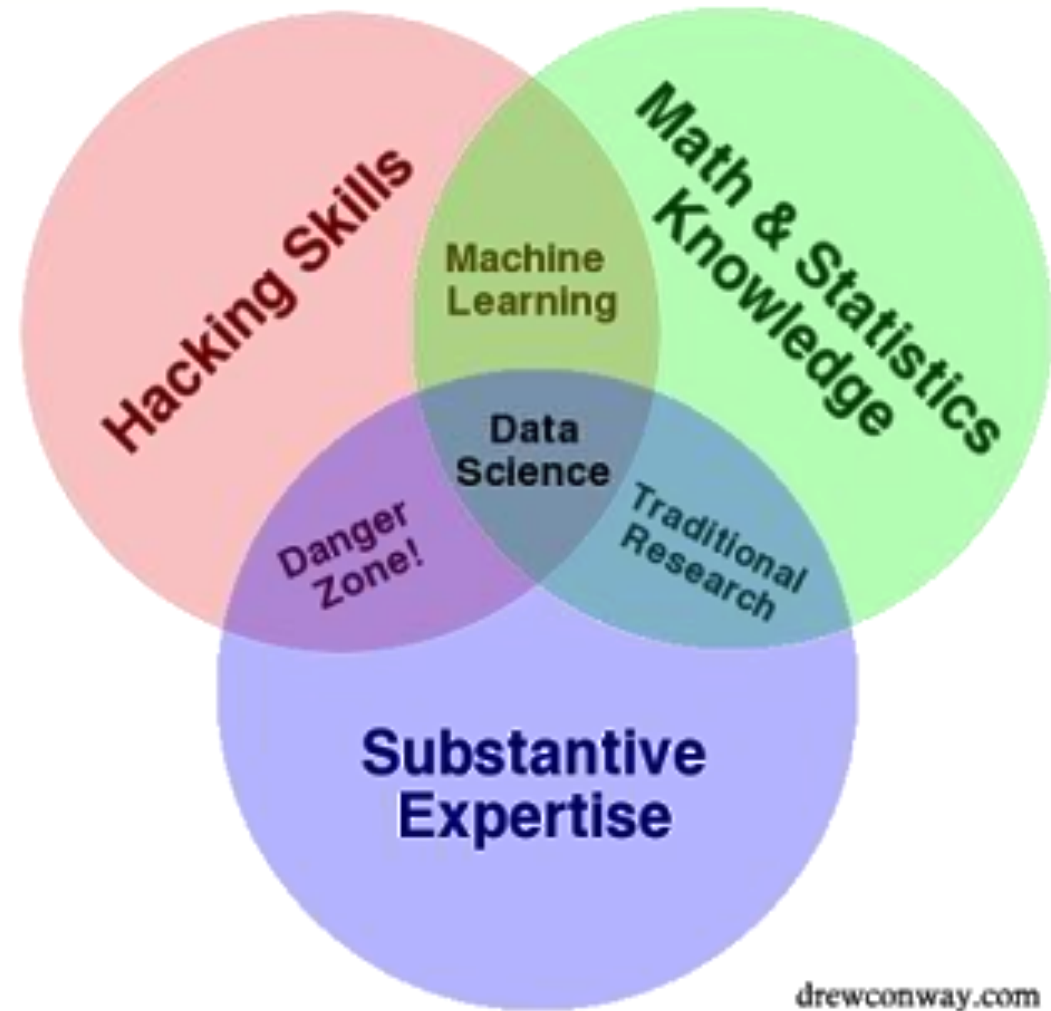
- 1 In 2014 a Facebook quiz invited users to find out their personality type
- 2 The app collected the data of those taking the quiz, but also recorded the public data of their friends
- 3 About 305,000 people installed the app, but it gathered information on up to 87 million people, according to Facebook
- 4 It is claimed at least some of the data was sold to Cambridge Analytica (CA) which used it to psychologically profile voters in the US
- 5 CA denies it broke any laws and says it did not use the data in the US presidential election
- 6 Facebook sends notices to users telling them whether their data was breached

CA denies any wrongdoing. Facebook has apologised to users and says a "breach of trust" has occurred.



Is data all you need?

Data science is a highly interdisciplinary activity concerned with gaining insights from often large and messy data which you sometimes have to collect, store and curate yourself. It is the application of data-centric, computer science, and inferential thinking to understand the world (science) and solve problems (engineering).





Example of questions we might try to answer with data science

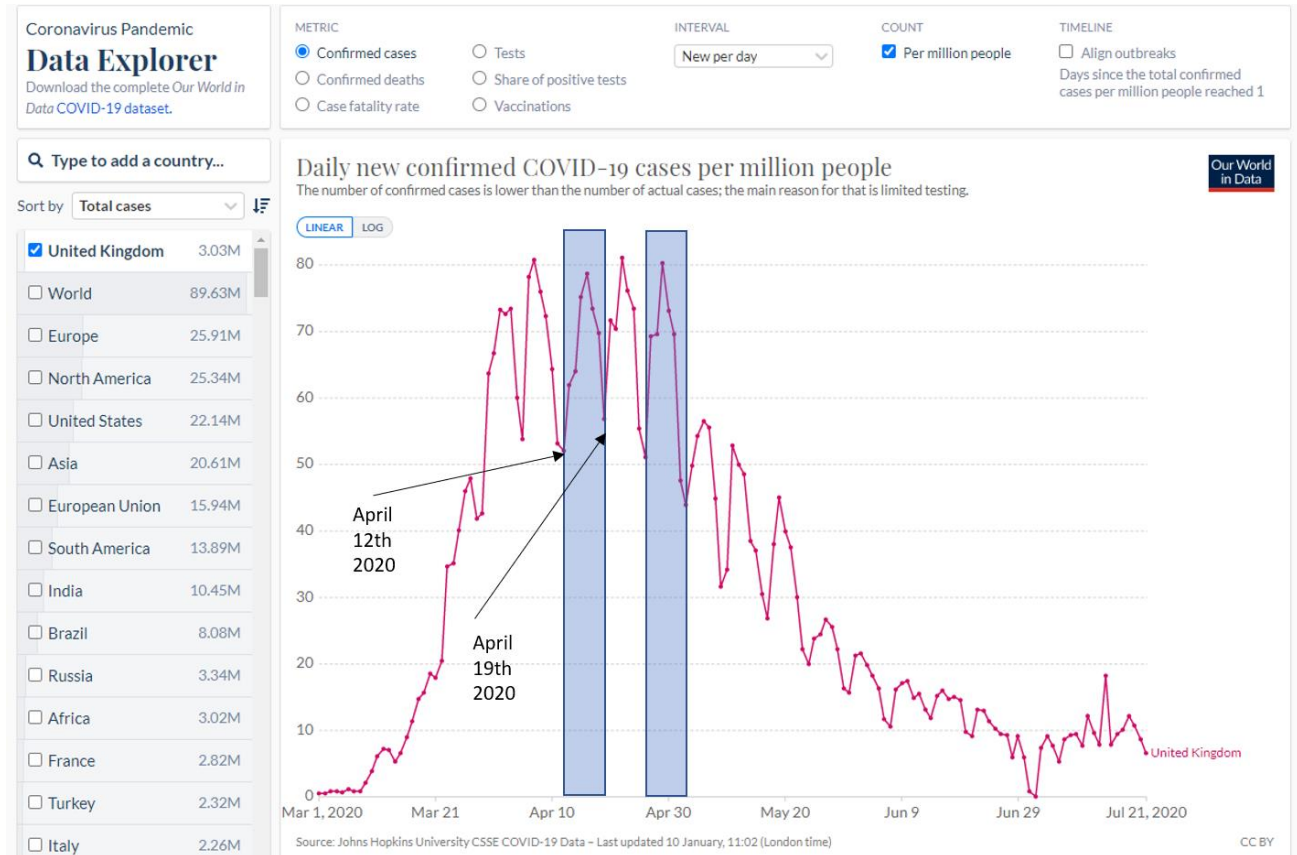
- How much exercise did you get in 2020?
- Is the use of the COMPAS algorithm for prison sentencing in the US fair?
- How will Brexit influence the UK economy?
- What should we eat to maximize our life expectancy?
- Who should get a COVID-19 vaccine first?
- Has the world gotten better or worse over the last 10 years (on average)?
- Will there be a forth wave of COVID19 in the UK?



Example: COVID-19 cases in the UK

Question: Why does the number of cases drop on certain days?

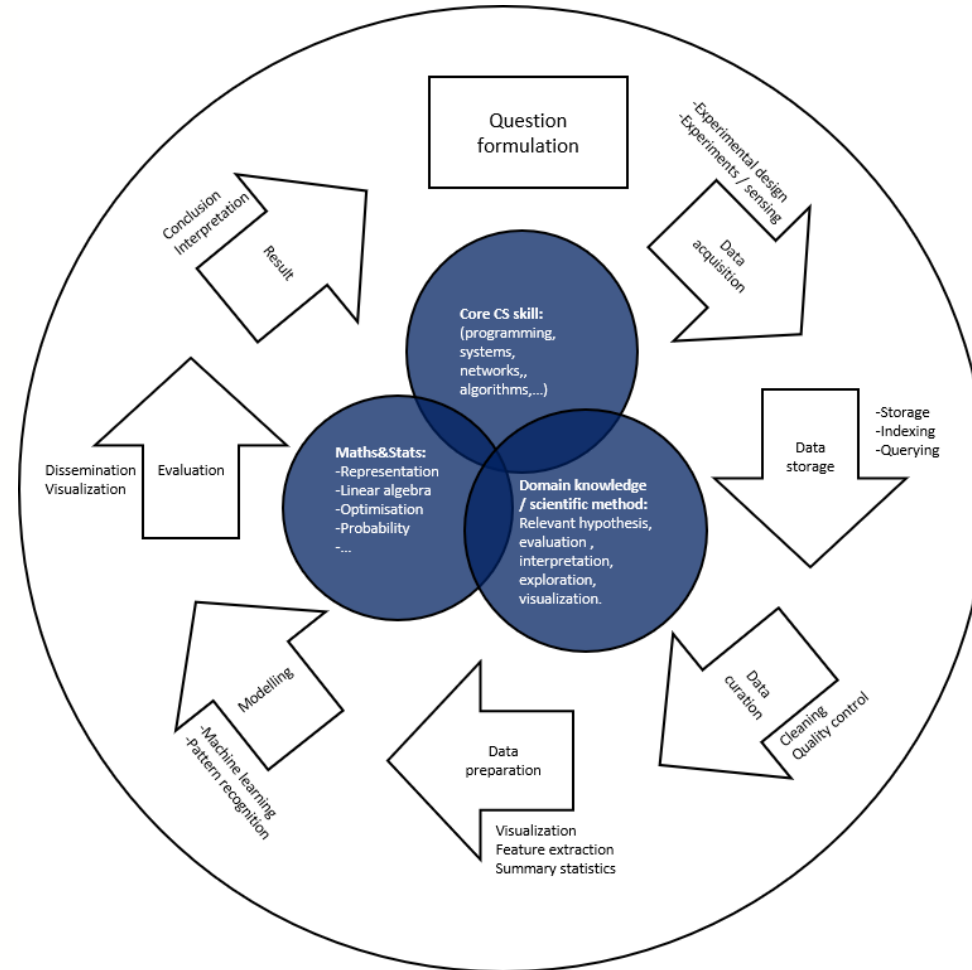
(consider what kind of information and knowledge you would need to answer this question)



*[Source: [Daily Covid cases from OutWorldInData, CC BY](<https://ourworldindata.org/coronavirus-data-explorer>)]*Interactive version



Data science and systems overview





About this course

Focus of this course

- *"How to represent data as numbers?"* --- Vector spaces.
- *"How to manipulate and transform data?"* --- Computational linear algebra (e.g. PCA/SVD).
- *"How to answer questions with data?"* --- Numerical optimisation.
- *"How to represent and reason with uncertainty?"* --- Probability & the basics and inference.
- *"How to store and access data?"* --- Data storage

Other skills

- Numpy (in detail) and Pandas (self-study / recommended material)
- Visualization
- Critical analysis and reflection
- Practical with a real-world data science case

The course is not: Machine learning, artificial intelligence, a "tools course" teaching you specific data science systems .



Practicalities

Credits and workload: - The course is a 15-credit course which amounts to about 150 hours of work.

Lectures (Thursdays) - In-person (in JMS building). Recordings of the live lecture will typically be provided on Moodle as Zoom/MS Streams

Lab sessions (Fridays)

- In Boyd Orr 1028
- Prepare labs in advance to the lab session!
- Peer support: Please be active in the MS Team groups and answer each other questions etc.

Assessment

- Labs (4 labs, each worth 8% of the final grade, recommended effort is 12 hours each) - deadlines posted on Moodle.
 - Labs are typically due 2/3 weeks after hand-out (see Moodle / LTC system for details).
- Weekly quizzes (10 quizzes, 6% in total, top 70% counts, i.e. you can drop two quizzes and still get top mark)
- Exam, written 2-hours in December.

Material. - The lecture notes (e.g. interactive notebooks, slides) with references to additional/supporting material. - Explore on your own! There are many excellent sources of information on the internet and we strongly encourage you to read outside the predefined curriculum.

A note on marking and effort: The suggested 12 hours for each assessed lab is the time we expect you to spend on the lab (effective work hours) to get the grade you would normally get in a similar course. An A1 (top grade) student should spend 12h getting an A1, a B2 (average) student should spend 12h getting a B2, etc.

A note on plagiarism: Don't do it! We automatically check all submissions against each other - and we are good at finding patterns. If you find yourself in a situation where copying other people's work is the only option then reach out to the teaching staff on the course and we can help/guide you.

A heads up: We will be monitoring and analysing the progress throughout the 11 weeks - we are data scientists after all! We might make changes to the schedule as we go along based on your feedback and our observations.



Feedback on your work

You will receive feedback on your work in a variety of ways in this module:

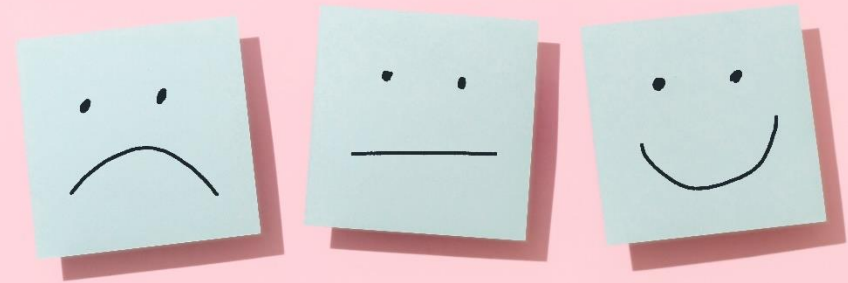
- The lab exercises are automarked based on running a range of tests on your code. The majority of those tests are visible: they will show you immediately if you've made a mistake.
- Additional hidden tests will be run on your work and the results (along with the mark) will be returned to you between 1 and 2 weeks after submission.
- One Lecturer and two Teaching Assistants will attend the weekly lab sessions and will be available to answer questions and provide advice on your work - **make the best use of this time!**
- The MSTeams channel is provided for discussions and questions about the course content.





Student feedback

- We monitor student feedback on the course in a variety of ways throughout the year:
- You will be asked to fill module feedback surveys twice in the term: at midpoint and towards the end. Please, complete them - they are an essential tool for us to monitor and improve the delivery of the course (and as you'll learn in this module, small samples are not great!).
- You can raise issues via the student representatives for discussion in the Staff-Student Liaison Committees.
- Don't hesitate to talk to us, during lectures, labs or via MSTeams.
- **We value your opinion!** We cannot change *what* you have to learn in this term (you'll need it in other modules next term!), but we are always looking to improve *how* we teach it.





Last year's feedback

Changes this year:

- **Slower pace:**

We have removed the more advanced databases content of the course, so that *Part 2: Optimisation* and *Part 3: Probabilities* are now taught over 3 weeks instead of 2. This should **reduce the pace of the course** to a more manageable level (note that the content is still, as intended, challenging!)

- **Fewer coursework:**

We have **reduced the number of assessed pieces of coursework** from 4 (at 6% each) to only 3 (at 8% each), to give you more time to complete each assignment (3 weeks instead of 2).

What you liked:

- Assignments are simulating and enacting
- course materials are good and professors explain them well.

What could be improved:

- Less homework or more duration time for it
- Maybe the pace of class can be slower. I feel like there is a lot of content in each class



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Questions?





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Thank you

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